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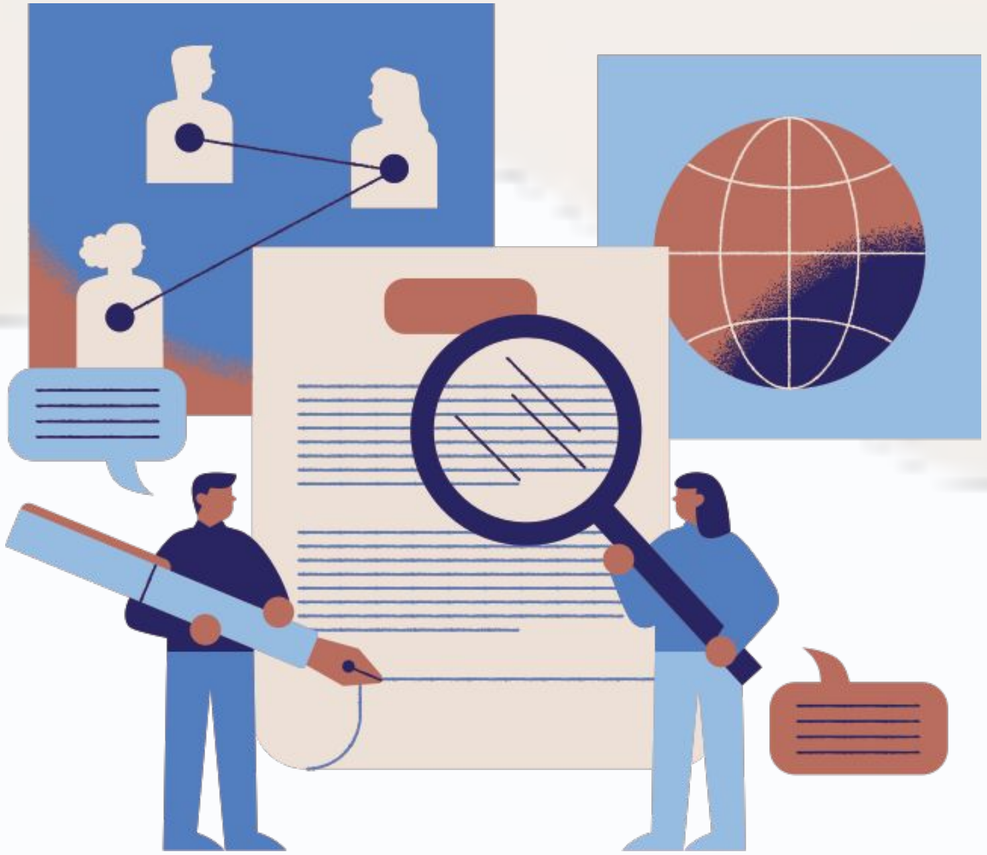
Sri Adichunchanagiri Shikshana Trust®

SJB INSTITUTE OF TECHNOLOGY

AN AUTONOMOUS INSTITUTE UNDER VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Course: Exploratory Data Analytics

Course Code: 23CSE422



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Introduction to EDA

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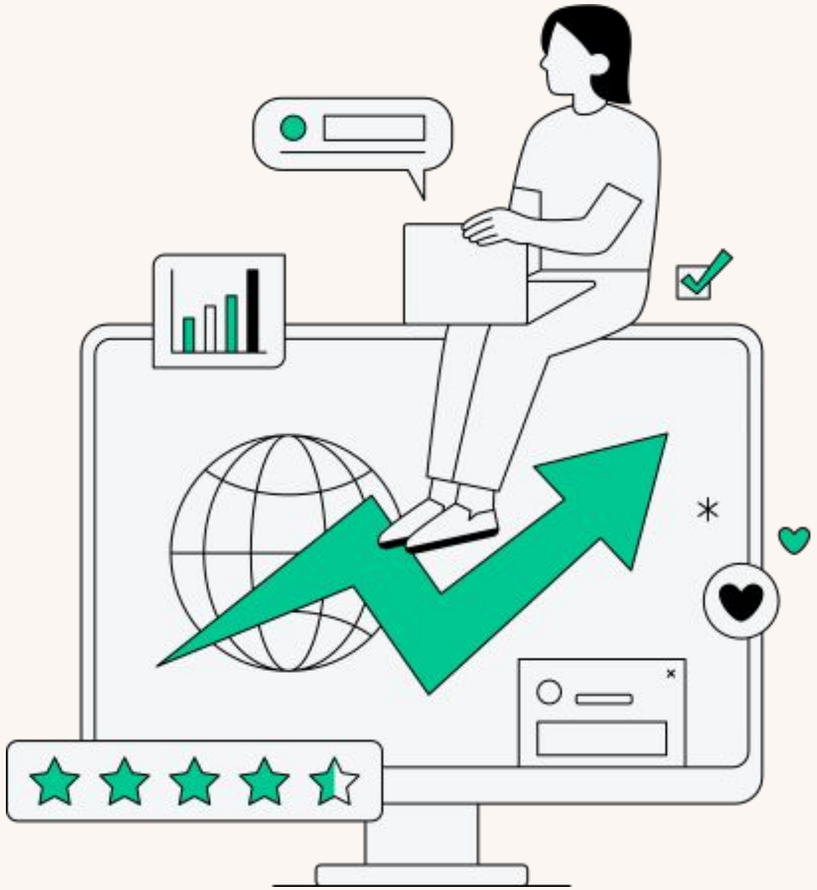


SCENARIO A

The Crowded Canteen

The college canteen sells out of Masala Dosa by 10 AM every day, but the sambar counter has leftovers at noon.

SCENARIO B

The Empty Classroom

Prof. Rao notices attendance drops every Monday at 8AM but is full on Fridays.

SCENARIO C

The Silent Bus Stop

A BMTC route may be cancelled due to low ridership — but commuters say the bus is always full in mornings.

SCENARIO D

The Viral Reel

A student's Instagram Reel about campus life got 50K views. The college wants to replicate this.

SCENARIO A

The Crowded Canteen - — Expected Data

The college canteen sells out of Masala Dosa by 10 AM every day, but the sambar counter has leftovers at noon.

No.	Data Item	How to Collect It
1	**Sell-out time of Masala Dosa each day**	Timestamp of last Dosa transaction in POS system or cashier log
2	**Number of Dosa prepared each morning**	Kitchen prep register or daily cook's log
3	**Number of Dosa sold per day**	Sales receipt count or POS transaction records
4	**Number of Sambar portions left over at noon**	End-of-service inventory count by canteen staff
5	**Peak footfall time at the canteen**	Manual headcount at entry or CCTV-based people counter
6	**Student visits 8–10 AM vs. 10 AM–12 PM**	Entry log or token system timestamps
7	**Student preference survey — top items bought**	Short Google Form or paper survey to students
8	**Price of Masala Dosa vs. Sambar per serving**	Canteen price board or billing records

SCENARIO A

Formal Definition + What is NOT Data

The Crowded Canteen

Formal Definition

Data is any recorded, retrievable fact about a real-world object, event, or phenomenon. In this scenario — the sell-out time is a recorded fact about an event. The portions prepared is a recorded fact about an object. The footfall count is a recorded fact about a phenomenon.

What is NOT data — Scenario A

Statement written by students	Why it is NOT data
"Masala Dosa is the most popular item"	Conclusion drawn from sales figures — not data itself
"The canteen is mismanaged"	Opinion or judgement — cannot be measured or recorded as fact
"Students love Dosa"	Interpretation of behaviour — not a recorded observation
"Management should make more Dosa"	Recommendation — comes after analysis, not before it
"Everyone rushes to canteen at 9 AM"	Generalisation — the measured headcount is the actual data

SCENARIO B

The Empty Classroom — Expected Data

Prof. Rao notices attendance drops every Monday at 8 AM but is full on Fridays.

No.	Data Item	How to Collect It
1	**Attendance count per student per day of week**	Biometric system or manual register
2	**Day-wise attendance % over one semester**	Calculate from register — present ÷ total students × 100
3	**Timetable slot — Monday 8 AM vs. Friday**	Department timetable document
4	**Subject scheduled at Monday 8 AM**	Timetable or academic calendar
5	**Student home distance — hostel vs. day-scholar**	Admission records or transport survey
6	**Monday public holidays in the semester**	Academic calendar — count Mondays that were holidays
7	**Student feedback rating for Monday 8 AM subject**	Anonymous end-of-term feedback form
8	**Whether Monday 8 AM has a test or deadline**	Faculty records or published course schedule

SCENARIO B

Formal Definition + What is NOT Data

The Empty Classroom

Formal Definition

Data is collected observations or measurements that represent the state of the real world at a specific point in time. Attendance count on a specific Monday is an observation captured at a specific point in time. Prof. Rao's pattern notice is NOT data — the actual Monday headcount figures are the data.

What is NOT data — Scenario B

Statement written by students	Why it is NOT data
"Students are lazy on Mondays"	Character judgement — no measurement supports this directly
"The 8 AM slot is too early"	Opinion — only valid as data if captured in a rated survey
"Attendance is bad this semester"	Vague generalisation — actual day-wise percentages are the data
"Friday is a more popular day"	Conclusion derived from counts — not the data itself
"Something happens every Monday"	Observation trigger — useful for forming a question, not data

SCENARIO C

The Silent Bus Stop — Expected Data

A BMTC route may be cancelled due to low ridership — but commuters say the bus is always full in mornings.

No.	Data Item	How to Collect It
1	**Passenger boarding count per trip (AM vs. PM)**	Conductor's daily trip sheet or automated ticketing machine
2	**Number of trips operated on this route per day**	BMTC route schedule and actual trip log
3	**Time of each trip — first, last, and peak hour**	BMTC timetable for this route
4	**Revenue collected per trip on this route**	Ticketing machine data or conductor's cash report
5	**Passenger count at stop during 7–9 AM**	Manual observer count or camera-based counting
6	**Number of passengers boarding at each stop**	Conductor's boarding log per stop
7	**Average bus occupancy as % of seat capacity**	Boarding count ÷ total seat capacity per trip
8	**Complaints or requests filed for this route**	BMTC grievance portal records

SCENARIO C

Formal Definition + What is NOT Data

The Silent Bus Stop

Formal Definition

Data is any raw, unprocessed fact captured from the real world that can serve as the basis for reasoning. This scenario presents a conflict between two data sources — the official ridership count and commuter reports. Both are data, measuring different slices of the same reality. The official count averages the full day. Commuters report the morning peak. Neither is wrong — they measure different things.

What is NOT data — Scenario C

Statement written by students	Why it is NOT data
"The bus is always full"	"Always" is a generalisation — boarding count per trip is the data
"Ridership seems low"	"Seems" is perception — the ticketing machine figures are the data
"This route is not profitable"	A financial conclusion — revenue and cost figures are the data
"The bus should not be cancelled"	Policy recommendation — comes after analysis, not before
"People depend on this route"	Assertion — passenger count and survey responses make this measurable

SCENARIO D

The Viral Reel — Expected Data

A student's Instagram Reel about campus life got 50K views. The college wants to replicate this.

No.	Data Item	How to Collect It
1	**Total view count of the Reel (50,000)**	Instagram Insights panel on the creator's account
2	**Average percentage of video watched per viewer**	Instagram Insights — Average watch time metric
3	**Number of shares and saves on the Reel**	Instagram Insights — shares and saves count
4	**Number of likes and comments**	Visible on post; detailed in Insights
5	**Time and day the Reel was posted**	Post timestamp visible on Instagram
6	**Account follower count at time of posting**	Creator's profile or archived screenshot
7	**Hashtags used in the caption**	Caption text of the post
8	**Duration of the Reel in seconds**	Visible on post metadata
9	**Audience age group and location of viewers**	Instagram Insights — audience demographic breakdown
10	**Replay and drop-off points in the video**	Instagram's video retention graph in Insights

SCENARIO D

Formal Definition + What is NOT Data

The Viral Reel

Formal Definition

Data is a recorded attribute or measurement that can be stored, retrieved, and compared across instances. This scenario introduces machine-generated data — every metric in Instagram Insights was recorded automatically without any human writing anything down. The view count, watch time, and retention graph were generated by the platform's systems as users interacted. The student did not collect this data — Instagram's servers did.

What is NOT data — Scenario D

Statement written by students	Why it is NOT data
"The Reel went viral"	"Viral" is a label assigned after the fact — view count is the data
"The content was creative"	Creativity is subjective — not measurable unless rated in a survey
"Students liked the vibe"	Vibe is unrecorded perception — likes and saves are the proxy data
"It was posted at the right time"	Conclusion — posting timestamp correlated with views is the data
"The video quality was good"	Subjective judgement — resolution, frame rate, bitrate are measurable

The Pattern Across All Four Scenarios

Every piece of 'not data' has a measurable fact hiding underneath it

Observation or Opinion

↓ Ask

Specific, Recordable, Retrievable Fact

= DATA

Scenario A

"Masala Dosa is popular"

→

Daily sell-out time + sales count

Scenario B

"Attendance is bad on Mondays"

→

Day-wise headcount register entry

Scenario C

"The bus is always full"

→

Boarding count per trip per stop

Scenario D

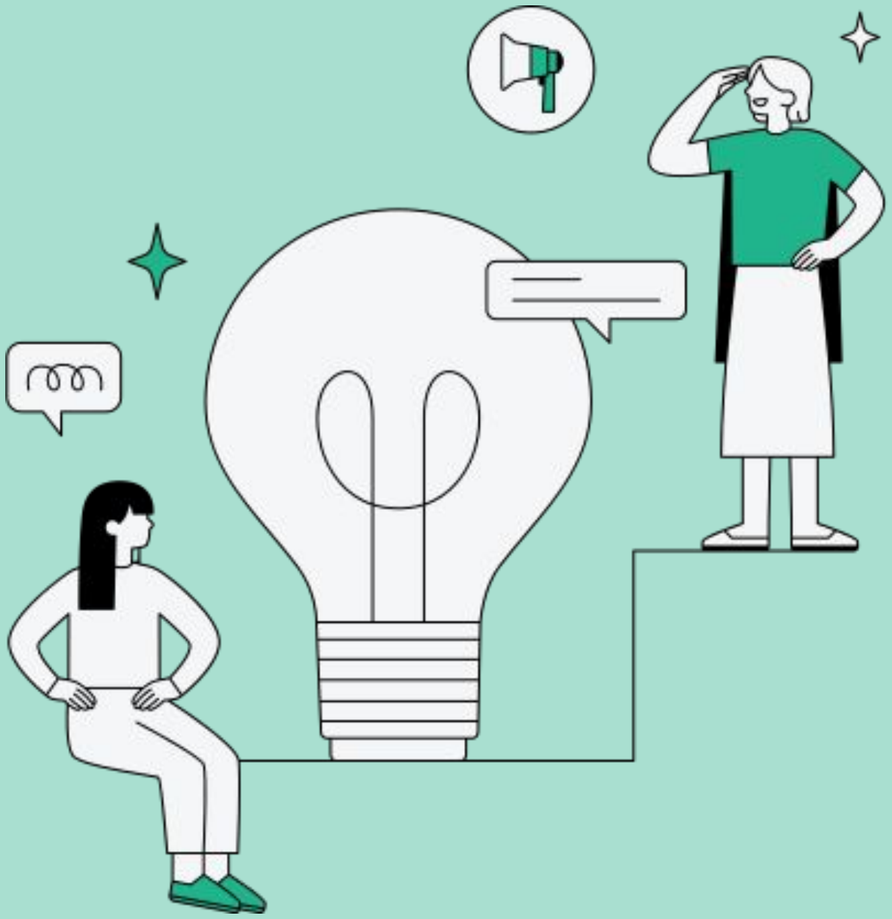
"The Reel went viral"

→

View count, watch time, share count

What is Data

- Definition:
  - Data is a collection of discrete objects, numbers, words, events, facts, measurements, or descriptions.
- Sources of Data:
  - Collected and stored during events or processes in various disciplines (e.g., biology, economics, engineering).
- Purpose of Data:
  - To extract useful information and generate knowledge.



Sources of Data

1️⃣ Primary Sources (First-hand Data)

This is data collected **directly** from the source for a specific purpose.

✅ **Examples:**

- Surveys & Questionnaires 📝 – Customer feedback collected by a company.
- Experiments 🧪 – Scientists measuring chemical reactions in a lab.
- Observations 👀 – A researcher noting traffic patterns in a city.
- Interviews 🎤 – A journalist collecting quotes from eyewitnesses.

2️⃣ Secondary Sources (Pre-collected Data)

This is data **already collected and published** by someone else.

✅ **Examples:**

- Government Reports 📊 – Census data, economic surveys.
- Research Papers 📚 – Studies published by universities.
- Company Databases 💾 – Sales reports, employee records.
- News Articles 📰 – Data compiled from different events.

3️⃣ Machine-Generated Data (Automated Data)

Data created by **computers, sensors, and devices** without human input.

✅ **Examples:**

- Web Analytics 🌐 – Google tracking website visitors.
- IoT Sensors 📡 – Smartwatches tracking heart rate.
- GPS & Location Data 📍 – Maps tracking real-time traffic.
- Automated Transactions 💰 – ATMs recording withdrawals.

4️⃣ Online & Social Media Data

Massive amounts of data are generated through digital platforms.

✅ **Examples:**

- Social Media Posts 📱 – Tweets, Facebook updates, YouTube comments.
- E-commerce Data 🛒 – Amazon tracking customer purchases.
- Search Engine Queries 🔍 – Google searches providing trends.
- Online Reviews ⭐ – Ratings and feedback on products.

5️⃣ Open Data Sources

Freely available datasets from public institutions and organizations.

✅ **Examples:**

- Government Portals 🌍 – Data from WHO, NASA, World Bank.
- Open-Source Datasets 📂 – Kaggle, UCI Machine Learning Repository.
- Weather & Climate Data 🌦️ – OpenWeather, NOAA records.
- Public APIs 🔗 – Wikipedia, Twitter API for extracting data.

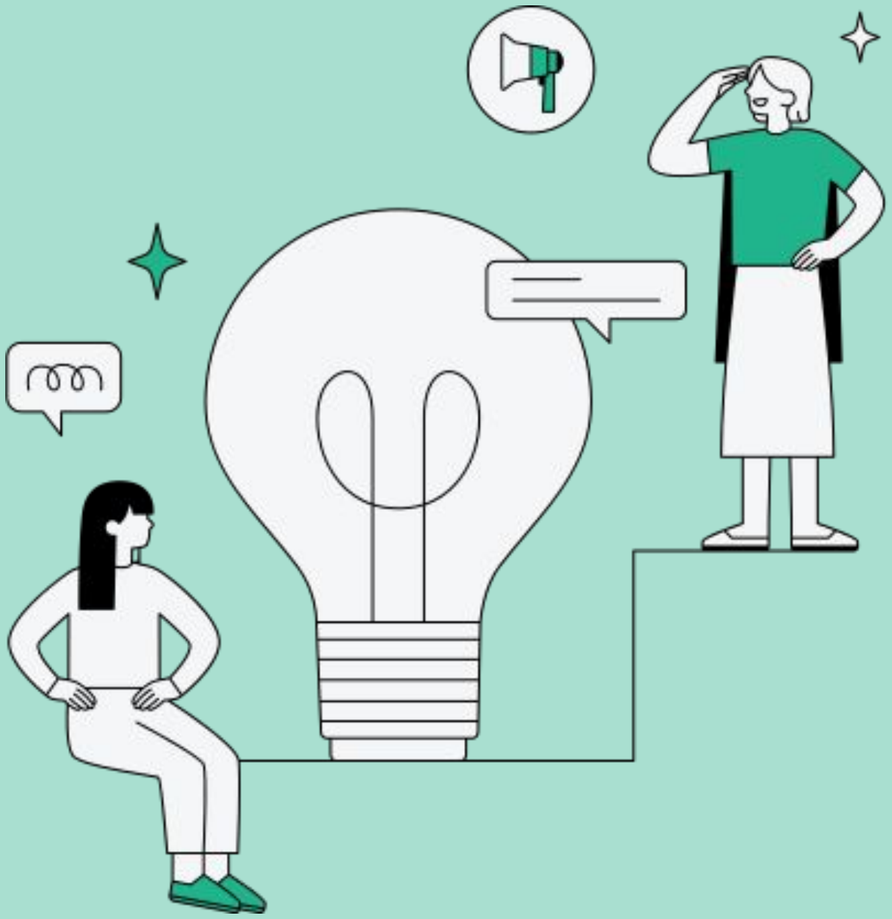
What is EDA?

Definition:

EDA is the process of examining datasets to

- **Discover patterns,**
- **Spot anomalies,**
- **Test hypotheses,**
- **Check assumptions**

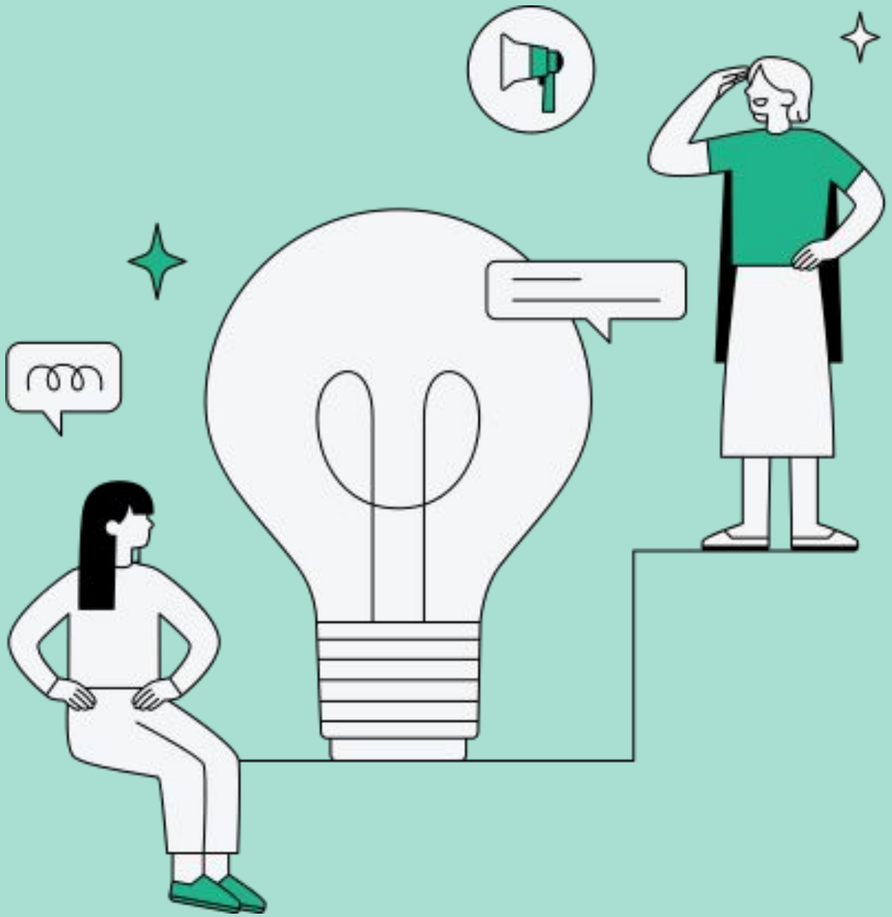
Using Statistical measures and visualizations.



Learning Goals

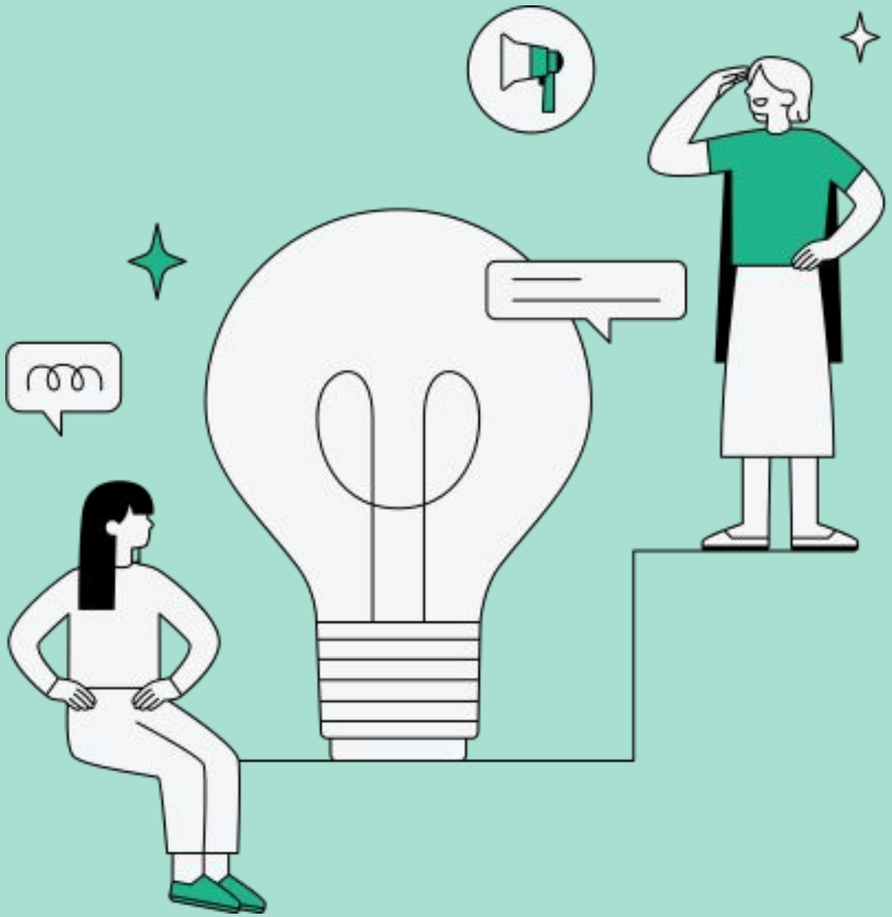
In this chapter, we will learn and revise the following topics:

- Understanding data science
- The significance of EDA
- Making sense of data
- Comparing EDA with classical and Bayesian analysis
- Getting started with EDA



Overview of EDA Stages

1. Data Requirements
2. Data Collection
3. Data Processing
4. Data Cleaning
5. Exploratory Data Analysis
6. Modeling and Algorithm Development
7. Data Product Creation
8. Communication



1. Data Requirements

- Sources of Data: Data can come from multiple sources like sensors, databases, or user interactions.
- Example: Application for monitoring dementia patients' sleep patterns:
- Data Required:
  - Sleep data
  - Heart rate
  - Electrodermal activity
  - User activity patterns
- Correct diagnosis requires all these data points to be accurately collected and stored.
- Categorizing Data:
  - Numerical Data: Continuous data like heart rate or sleep duration.
  - Categorical Data: Data like activity patterns or types of sleep stages.
- Data Storage and Dissemination:
  - Define storage formats (e.g., tables, databases).
  - Ensure accessibility and proper dissemination for analysis.

2. Data Collection

- Multiple Sources: Data is gathered from various objects and events, such as sensors, devices, and user interactions.
- Correct Storage Format:
  - Data must be stored in the appropriate format for effective analysis (e.g., CSV, JSON, SQL databases).
  - Formats depend on the type of data and tools used for analysis.
- Transfer to IT Personnel:
  - Collected data must be transferred to the right personnel for further processing and handling.
  - Ensures proper data management and security during storage and transit.
- Types of Tools Used for Data Collection
  - Sensors:
    - IoT devices, wearables, environmental sensors
  - Storage Tools:
    - Databases (SQL/NoSQL), cloud storage systems, local storage

3. Data Processing

- Preprocessing:
  - The initial step in preparing data for analysis.
  - Ensures data is organized, clean, and ready for exploration.
- Importance of Pre-curation:
  - Reduces inconsistencies and errors in data.
  - Enhances the quality and reliability of analysis outcomes.
- Common Tasks in Data Processing
  - Exporting the Dataset:
    - Ensure data is exported correctly from its source (e.g., database, API, files).
  - Organizing Data into Tables:
    - Place data in appropriate tables or data frames for easy access and analysis.
  - Structuring the Data:
    - Arrange data in a logical and analyzable format (e.g., rows for records, columns for attributes).
  - Exporting in the Correct Format:
    - Save the processed data in widely used formats such as CSV, Excel, or JSON for compatibility with analysis tools.

4. Data Cleaning

- What is Data Cleaning?
  - The process of transforming preprocessed data to prepare it for detailed analysis.
  - Ensures the dataset is complete, accurate, and free from errors.
- Key Tasks in Data Cleaning
  - Incompleteness Check:
    - Identify and address missing values in the dataset.
  - Duplicates Check:
    - Detect and remove redundant records.
  - Error Check:
    - Spot inaccuracies or inconsistencies in the data.
  - Missing Value Handling:
    - Fill in or impute missing values to maintain dataset integrity.
- Analytical Techniques for Identifying Anomalies
  - Use statistical and visualization tools to detect data issues like outliers, inconsistencies, or gaps.
- Example of Data Cleaning
  - Outlier Detection: Apply quantitative methods (e.g., z-score, IQR) to identify and handle extreme values that may skew analysis.

23CSE422 — EXPLORATORY DATA ANALYTICS

Data Cleaning

Understanding Missing Data, Dirty Data & Outliers

Yesterday — Activity : The Messy Data Clinic | Today — Stage 4 of the EDA Pipeline: Data Cleaning

MISSING

DIRTY

OUTLIERS

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EDA PIPELINE

Where Does Data Cleaning Sit?

Stage 4 of 8 — and the most time-consuming stage in real projects

1	2	3	**4**	5	6	7	8
Requirements	Collection	Processing	Cleaning	Analysis	Modelling	Data Products	Communication

GIGO Principle

Garbage In, Garbage Out — no amount of sophisticated modelling can fix an analysis built on unclean data. Stage 4 exists to prevent GIGO.

Missing values

Detect and decide: re-collect, impute, or flag

Dirty / Invalid

Identify wrongly formatted or impossible values

Outliers

Flag extreme values and decide case by case

RECALL

Yesterday's Dataset — Class Health Check

There are 8 problems across 10 rows. Let's classify them formally.

#	Name	Age	Height	Weight	Blood	Score
1	Aarav	19	172	68	O+	74
2	Bhavna	20	158	—	B+	81
3	Chetan	21	312	72	A+	66
4	Deepa	19	161	54	AB-	89
5	Esha	20	165	59	—	77
6	Farhan	22	175	74	XY	83
7	Gauri	19	163	51	O-	180
8	Harsh	21	170	65	A-	—
9	Isha	20	155	4	B-	71
10	Jai	twenty	168	60	O+	79

8 problems total — how many did you find?

O — Outlier — present but extreme

M — Missing — absent or blank

D — Dirty — invalid format or type

MISSING DATA

What Is Missing Data?

A value that should exist in a cell but does not

Missing data occurs when no value is stored for a variable in a particular observation. The cell exists. The row exists. But the value was never recorded, was lost, or was deliberately withheld.

MCAR

Missing Completely At Random

The probability of being missing is the same for all observations. Example: a scale malfunctions randomly and misses Bhavna's weight. The missing value has nothing to do with Bhavna or her weight.

Least problematic. Safe to impute with mean/median.

MAR

Missing At Random

Missingness depends on other observed variables, but not on the missing value itself. Example: older students are less likely to report pain levels. Age predicts missingness — but knowing age lets you adjust.

Manageable. Use group-based imputation.

MNAR

Missing Not At Random

Missingness depends on the unobserved value itself. Example: students with very low exam scores are more likely to leave the score blank. The blank is not random — it is informative.

Most dangerous. Imputing without care introduces systematic bias.

MISSING DATA

The 3 Missing Values — What They Each Mean

Same label M, but very different situations and very different responses

Row 2 — Weight (Bhavna)

MCAR (likely)

Bhavna's weight is blank. No indication it is related to her actual weight or any other recorded variable.

Options:

1. Re-collect from source — original measurement card
2. Impute with class mean (≈59 kg) or median
3. Mark as NaN and exclude from weight-specific analysis

Never: Do not impute without documenting the method chosen.

Row 5 — Blood Type (Esha)

MNAR (possible)

Blood type is blank. Could be refused (private), could be unknown to student, or simply not recorded. Categorical — cannot average.

Options:

1. Re-collect from original health record
2. Label as 'Unknown' category
3. Exclude Esha from blood-type-specific analysis only

Never: Never impute a blood type with the mode — blood type is medically critical personal data.

Row 8 — Exam Score (Harsh)

MNAR (likely)

Score is blank. Harsh attended — signature on sheet. Paper may be lost, or he was absent for the exam only. The blank may carry information.

Options:

1. Investigate: check the physical paper, check absence records
2. If paper is lost: escalate to faculty — administrative resolution needed
3. Do not impute. Assigning a score to a student who may not have sat the exam is a data integrity violation.

Never: Do not impute with class mean — this assigns a score Harsh did not earn.

DIRTY DATA

What Is Dirty Data?

Values that are present but wrong — wrong type, wrong format, or impossible

Dirty data is present in the dataset but violates the rules of the variable it occupies — wrong data type, impossible value, wrong format, or value that belongs to a different column.

A
a

Wrong Data Type

Age = 'twenty' → text in a numeric column

The entire Age column is read as text by any analysis tool. All numeric operations on that column fail silently or return errors.

?

Invalid Category

Blood Type = 'XY' → not a valid blood group

Cannot be imputed. The value is present but meaningless within the domain. Must be investigated and verified.

≠

Format Inconsistency

Date entered as '12-03-26', '12/03/2026', 'Mar 12'

Three formats for the same date. Any date comparison or sort will fail unless standardised first.

↔

Cross-Column Error

Gender code 'XY' entered into Blood Type column

Data belongs to the wrong variable. Suggests a data entry process failure — one that may have occurred across many rows.

DIRTY DATA

The 2 Dirty Values — Diagnosis & Fix

Row 6 and Row 10 — different dirty types, different responses

Row 6 — Blood Type: 'XY'

Invalid category

Valid blood types: A, B, AB, O (with + or −)
'XY' does not exist in this domain.

Most likely cause:

XY is a chromosomal sex designation, not a blood type. This looks like gender notation accidentally entered in the Blood Type column — a cross-column data entry error.

What to do:

1. Verify against original health record
2. If unresolvable, treat as Missing — label Unknown
3. Document the investigation and outcome
4. Review if other rows show the same pattern

Never: Replace with a guessed blood type.

Row 10 — Age: 'twenty'

Wrong data type

Age column expects an integer.
'twenty' is a word — wrong data type.

Why this is dangerous:

In any analysis tool — Python, Excel, R — one text entry in a numeric column forces the entire column to be treated as a text (string) column. Operations like mean, median, standard deviation, and comparison all fail silently or return errors for the entire column.

What to do:

1. The value is recoverable — convert 'twenty' to 20
2. Document the conversion: 'twenty → 20, text→int'
3. Investigate the data entry process: is this a form validation issue? Could there be more?

Never: Fix silently without documenting the change.

OUTLIERS

What Is an Outlier?

A value that is present, plausible in format, but statistically extreme

An outlier is an observation that lies an abnormal distance from the other values in the dataset. It is technically valid data — the cell is not empty, the type is correct — but it is so extreme that it warrants investigation before inclusion in analysis.

Two ways to detect outliers:

Z-Score Method

$$z = (x - \mu) / \sigma$$

x = value, μ = mean, σ = std deviation

A z-score beyond ±3 means the value is more than 3 standard deviations from the mean. Statistically, fewer than 0.3% of values in a normal distribution fall outside this range.

Applied to Row 9: weight = 4 kg, class mean ≈ 62 kg, $\sigma \approx 8$ kg → $z = (4 - 62) / 8 = -7.25$ → Extreme outlier.

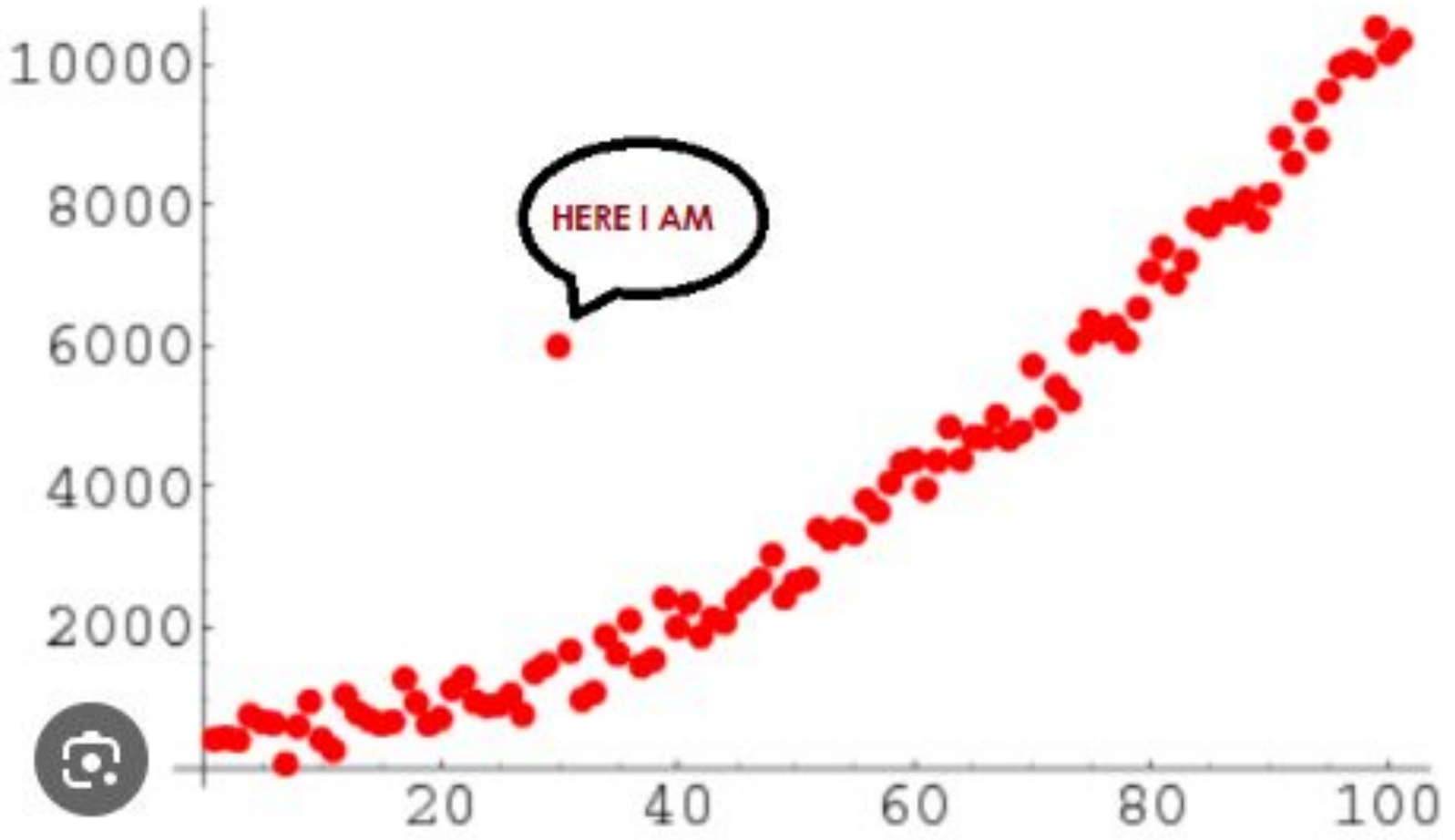
IQR Method

$$IQR = Q3 - Q1$$

Outlier if $< Q1 - 1.5 \times IQR$ or $> Q3 + 1.5 \times IQR$

The IQR method uses the middle 50% of data (Q1 to Q3) to define a normal range. Values outside 1.5× the IQR below Q1 or above Q3 are flagged. More robust than z-score for non-normal distributions.

Does not require the data to be normally distributed — preferred when distribution is unknown or skewed.


10000
8000
6000
4000
2000
20
40
60
80
100
HERE I AM

OUTLIERS

The 3 Outliers — What Each One Tells Us

Height 312 cm · Exam Score 180 · Weight 4 kg

Row 3 — Height: 312 cm

z ≈ +31 — physically impossible

The tallest human ever recorded was 272 cm. 312 cm cannot be a real height for a college student.

Most likely cause:

Almost certainly a data entry error — transposed digits. Likely 132 cm, 172 cm, or 213 cm.

What to do:

1. Check the original measurement record
2. If unresolvable: flag as suspect, do not delete
3. Report with and without this value to show impact

Row 7 — Exam Score: 180

Impossible — max is 100

The exam is scored out of 100. A value of 180 cannot exist on this scale.

Most likely cause:

Most likely 80 with an extra digit, or 18 misread. Could be a wrong scale (out of 200 entered as-is).

What to do:

1. Check original answer sheet or mark ledger
2. If it was out of 200: the entire column scale may be wrong
3. Confirm the maximum possible score before any fix

Row 9 — Weight: 4 kg

z ≈ −7.25 — extreme

4 kg is the weight of a newborn infant. No college student weighs 4 kg.

Most likely cause:

A missing zero. Most likely 40 kg or 64 kg. The digit '4' was entered without the second digit.

What to do:

1. Check the original measurement record
2. If unresolvable: do not impute — flag as suspect
3. Document: 'value 4 kg flagged as likely entry error, original unverifiable'

OUTLIERS

Real Outlier vs. Data Entry Error — The Critical Distinction

Not every outlier is a mistake. Not every extreme value should be removed.

Real Outlier — Keep It

- The value is extreme but physically possible
- It represents a genuine observation about a real entity
- Removing it would hide important information
- Example: a student scored 4/100 on an exam — extremely low, but real. Removing it hides a student who needs help.
- Example: a patient weighs 38 kg due to a medical condition. Removing this hides a medically significant case.

Response: Keep. Analyse with and without. Document the decision and its effect on results.

Entry Error — Investigate First

- The value is physically impossible (312 cm height)
- The value violates the scale of the instrument (score > 100)
- The value is a clear transposition or typo (4 kg vs 40 kg)
- Example: height 312 cm — tallest recorded human is 272 cm. This is not a real observation.
- Example: exam score 180 — maximum is 100. This cannot be a correct recording.

Response: Investigate. Verify against source. Correct if verifiable. Flag if not. Never delete silently.

5. Exploratory Data Analysis (EDA)

What is EDA?

- A crucial stage in data analysis where the dataset is explored to understand its underlying patterns and insights.
- Involves identifying trends, spotting anomalies, testing hypotheses, and validating assumptions.

Key Aspects of EDA

- Understanding the Data:
  - Gain a clear picture of the dataset's structure, variables, and relationships.
- Data Transformation:
  - Apply necessary techniques (e.g., normalization, scaling) to prepare data for analysis.
- Statistical Measures:
  - Use descriptive statistics to summarize data (e.g., mean, median, mode, variance).
- Why EDA Matters
  - Helps in uncovering hidden patterns and insights.
  - Assists in feature selection for modeling.
  - Provides a foundation for building robust and accurate data models.

6. Modeling and Algorithm

What is Modeling?

- Modeling involves creating mathematical formulas to describe relationships between variables.
- These relationships can show correlation or causation.

Key Concepts

1. Independent and Dependent Variables:
    - Independent Variable: Input factor influencing the outcome.
    - Dependent Variable: Outcome influenced by independent variables.
    - Example:
        - Total Price = Unit Price × Quantity.
        - Total Price = Dependent Variable.
        - Unit Price and Quantity = Independent Variables.
2. The Judd Model:
    - Data = Model + Error.
    - Emphasizes that models include inherent errors or uncertainties.

- Inferential Statistics in Modeling
    - Quantifies relationships between variables.
    - Example: Regression Analysis
        - Helps predict dependent variable outcomes based on independent variables.

7. Data Product

What is a Data Product?

- A data product is a computer software system that:
  - Uses data as input.
  - Produces output based on analysis.
  - Provides feedback to control or improve the environment.
- Key Characteristics:
  - Built upon models developed during data analysis.
  - Designed to automate decision-making or improve user experience.
- Example:
  - Recommendation Model:
    - Inputs: User purchase history.
    - Output: Suggests related items the user is likely to buy.
- Purpose of Data Products:
  - Enhance business intelligence.
  - Drive user engagement and satisfaction.
  - Provide actionable insights.

8. Communication

What is Communication in EDA?

- The stage where results from exploratory data analysis are shared with stakeholders.
- Helps in converting analysis into actionable business intelligence.

Key Aspect:

- Data Visualization:
  - The primary tool for communicating findings.
  - Uses visual aids to present analyzed data clearly and effectively.

Common Visualization Techniques:

- Tables
- Charts
- Summary Diagrams
- Bar Charts

Purpose of Visualization:

- Simplifies complex data for easier interpretation.
- Provides clarity and insights to drive informed decision-making.

Significance of Exploratory Data Analysis (EDA)

What is EDA?

- Exploratory Data Analysis (EDA) is the initial phase in the data mining process.
- It focuses on exploring and understanding datasets before applying more complex statistical models.
- EDA is crucial for generating hypotheses and visualizing the data to grasp patterns, trends, and relationships within the data.
- It serves as a preliminary investigation to decide which modeling techniques or methods are appropriate for the data.

Why is EDA Significant?

- Understanding Data:
  - Without proper exploration, making sense of large datasets with numerous variables becomes difficult.
  - EDA helps uncover hidden insights about the data, including distributions, patterns, and correlations.
  - By performing EDA, we get a clear picture of the ground truth of the data, enabling informed decision-making.
- Data-Driven Decision Making:
  - EDA is essential for making well-informed, data-driven decisions in various fields such as business, healthcare, and finance.
  - By using EDA techniques, we ensure that the dataset is properly understood and processed before moving forward in the analysis.
- Hypothesis Creation:
  - It allows us to form hypotheses or assumptions based on initial findings, which can later be tested with more sophisticated modeling.
  - Hypotheses generated in this phase serve as a foundation for further data mining and predictive analysis.

Significance of Exploratory Data Analysis (EDA)

Summarizing Data:

- Descriptive statistics like mean, median, mode, variance, and standard deviation are used to summarize the key characteristics of the data.
- Summarization helps to detect any outliers or anomalies early on, aiding in further cleaning or transformation tasks.

Statistical Analysis:

- Basic statistical tests and analysis, such as hypothesis testing, correlation analysis, and regression analysis, are conducted during EDA.
- These tests help to uncover any statistical relationships between variables, supporting further hypotheses for deeper analysis.

Data Visualization:

- Visualization techniques such as bar charts, histograms, box plots, scatter plots, and heatmaps help illustrate the data's distributions, relationships, and patterns.
- Visualization makes it easier to identify trends, clusters, and outliers, which may be difficult to detect through raw data alone.

Significance of Exploratory Data Analysis (EDA)

Tools for EDA

- Python provides powerful tools for performing EDA:
  - pandas: A library used for data summarization (mean, median, standard deviation, etc.), and manipulation, such as data cleaning and transformation.
  - scipy: A library that supports statistical analysis, hypothesis testing, and more advanced statistical functions.
  - matplotlib & plotly: Libraries that help create a variety of visualizations to reveal the underlying structure of data, such as charts, graphs, and plots.

Steps in EDA

4 Different steps involved in data analysis are:

1. Problem Definition
2. Data preparation
3. Data analysis
4. Development and representation of the results

1. Problem Definition: Define the business problem to solve.
    - Define the main objective of the analysis.
    - Identify deliverables.
    - Outline roles and responsibilities.
    - Assess current status of the data.
    - Set a timetable and perform cost/benefit analysis.

   Purpose: The problem definition drives the execution plan for data analysis.

2. Data preparation: Prepare the dataset for analysis.
    - Define data sources and schema.
    - Understand the characteristics of the data.
    - Clean the dataset (remove irrelevant data, handle missing values).
    - Transform and divide data into required chunks.

   Purpose: Ensure data is well-organized and ready for analysis.

Steps in EDA

3. Development and representation of the results:

Analyze the data using various techniques.

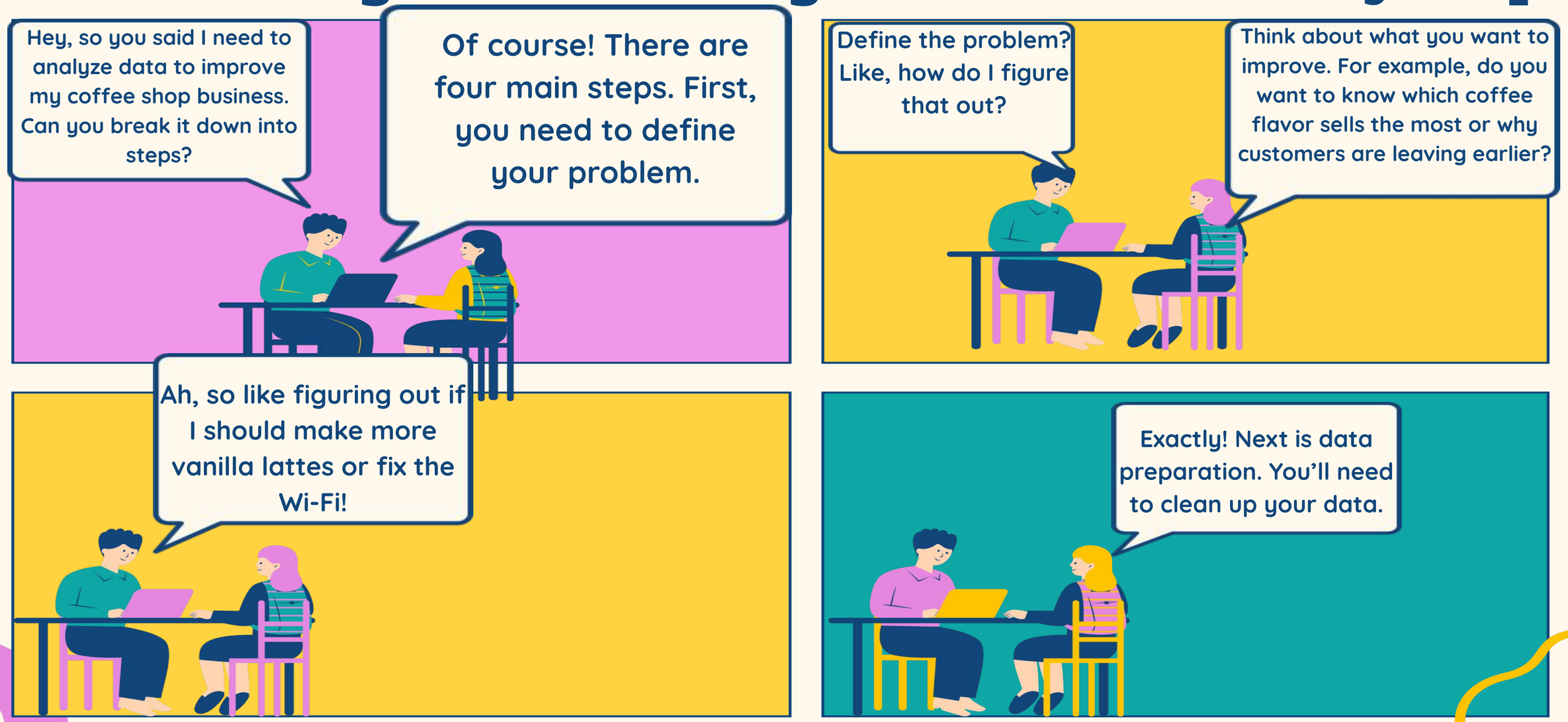
- Summarize the data.
- Identify correlations and relationships.
- Develop and evaluate predictive models.
- Calculate model accuracies.

- Techniques used for data summarization to Extract meaningful insights and patterns:
  - Descriptive statistics, correlation, inferential statistics.
  - Mathematical models, grouping, and searching.

4. Development & Representation of Results:

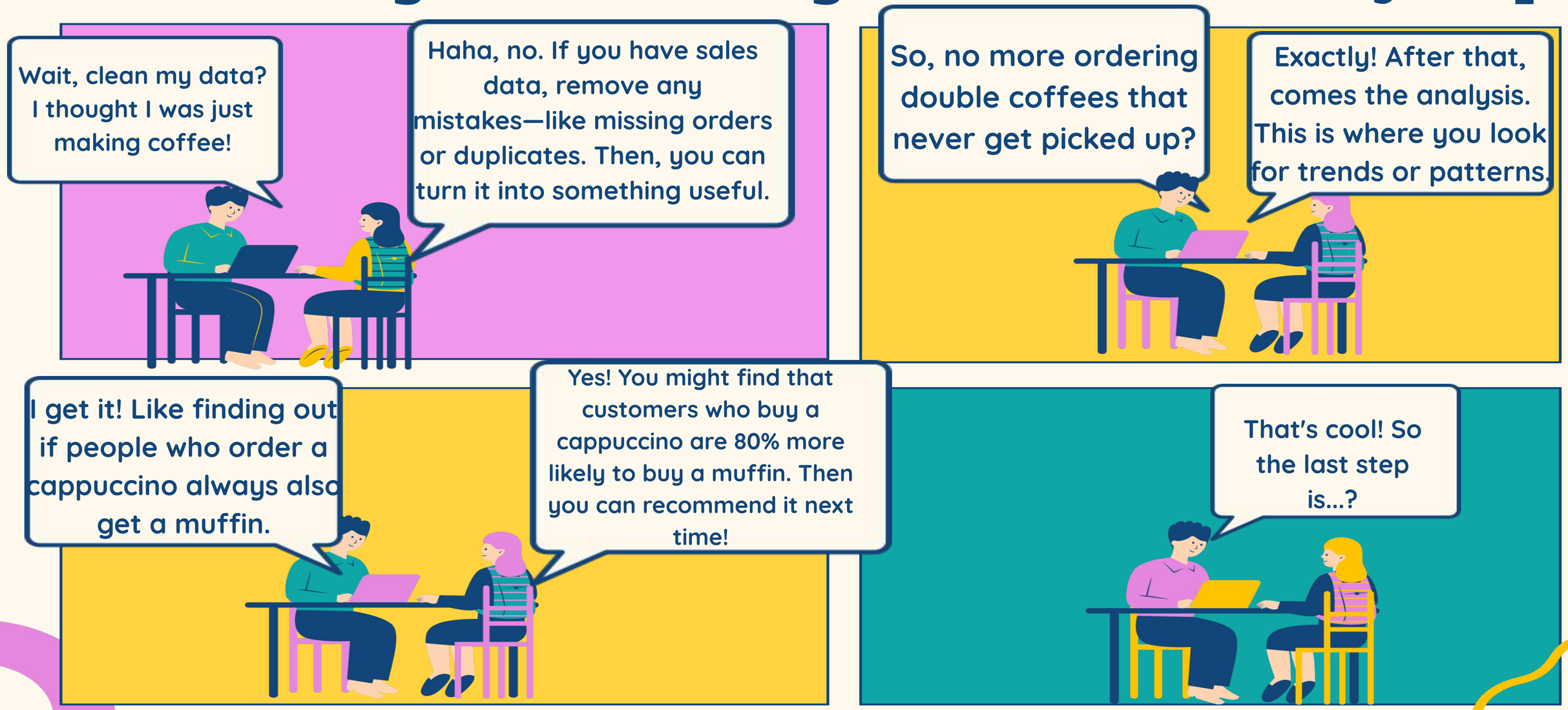
- Present the findings in an understandable format for stakeholders.
- Key Tasks:
  - Create visualizations: graphs, tables, maps, diagrams.
  - Communicate results effectively for decision-making.
- Types of Graphical Representations used to ensure the results are interpretable and actionable:
  - Scatter plots, histograms, box plots, residual plots.

Caffeinating Data: Brewing Success with Every Step

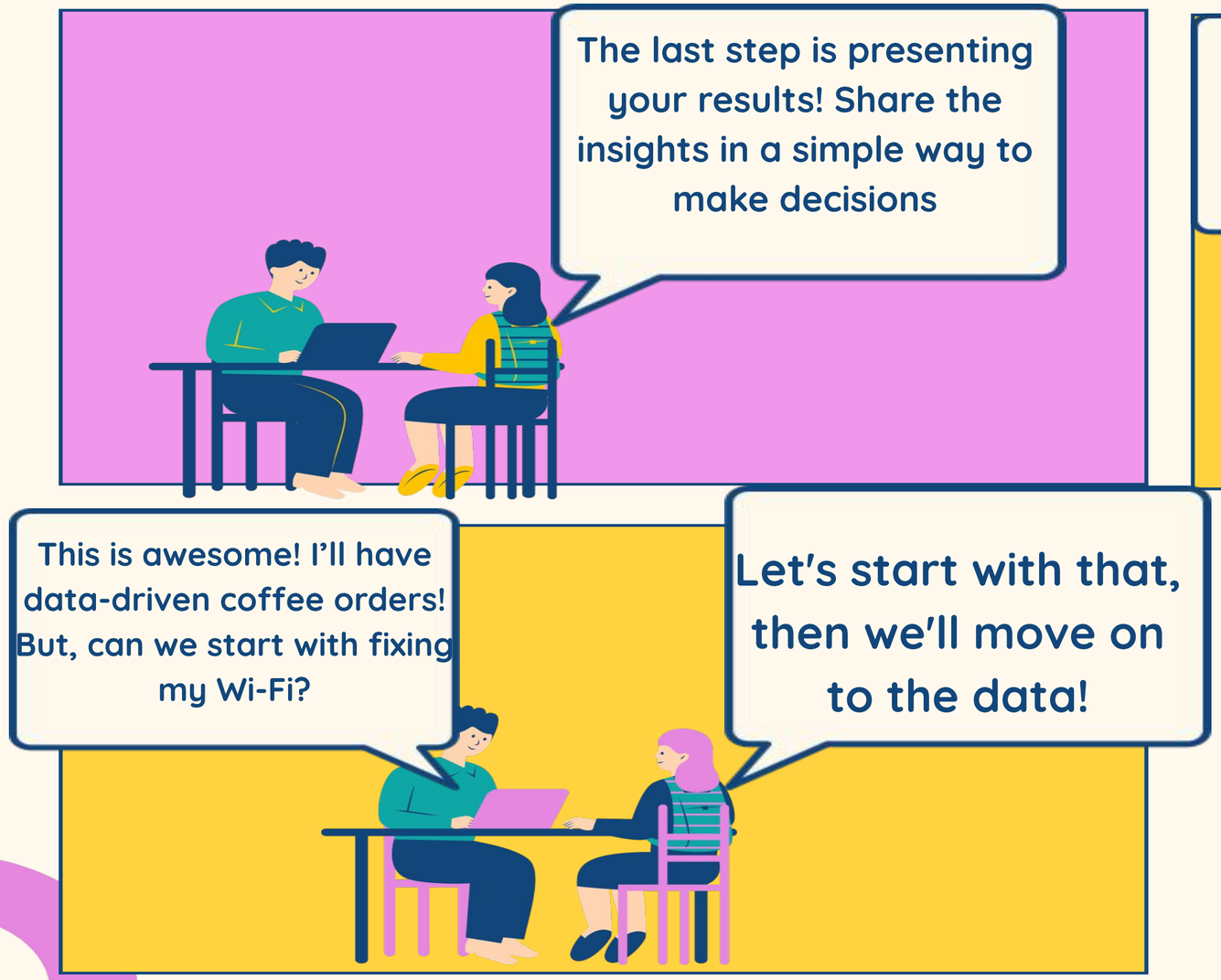


INSPIRED BY mathspace

Caffeinating Data: Brewing Success with Every Step



Caffeinating Data: Brewing Success with Every Step



Making Sense of Data

- Importance of Identifying Data Types:
  - Different disciplines store and analyze data for different purposes:
    - Medical researchers: Patients' data.
    - Universities: Students' and teachers' data.
    - Real estate: House and building data.
  - Understanding the type of data is critical for proper analysis.
- Dataset Structure:
  - A dataset consists of multiple observations about specific objects or entities.
  - Example: A hospital dataset containing patient information.
  - Each observation has several variables or features that describe the object.
  - Example: A patient described by Patient_ID, Name, Address, DOB, Email, Weight, Gender.
- Example of Patient Data:
  - Patient Information:
    - PATIENT_ID: Unique identifier for the patient.
    - Name, Address, Date of Birth (DOB), Email, Gender, Weight: Descriptive variables.
- Example record:

"PATIENT_ID = 1001, Name = Yoshmi Mukhiya, Address = Mannsverk 61, 5094, Bergen, Norway, DOB = 10th July 2018, Email = yoshmimukhiya@gmail.com, Weight = 10, Gender = Female"

Making Sense of Data

Dataset Table Example:

- The dataset is stored in tables/schema in database management systems.
- Example Table Structure for Storing Patient Information:

PATIENT_ID	NAME	ADDRESS	DOB	EMAIL	GENDER	WEIGHT
001	Suresh Kumar	Mannsverk, 61	30.12.1989	skmu@hvl.no	Male	68
002	Yoshmi Mukhiya	Mannsverk 61, Bergen, Norway	10.07.2018	yoshmimukhiya@gmail.com	Female	1
003	Anju Mukhiya	Mannsverk 61, Bergen	10.12.1997	anjumukhiya@gmail.com	Female	24
004	Asha Gaire	Butwal, Nepal	30.11.1990	aasha.gaire@gmail.com	Female	23
005	Ola Nordmann	Danmark, Sweden	12.12.1789	ola@gmail.com	Male	75

Key Components of a Dataset:

- Observations: Rows in the dataset (e.g., each patient record).
- Variables: Columns describing the data (e.g., Patient ID, Name, Weight).

Data Types:

- Numerical Data: Quantitative variables like weight, age, etc.
- Categorical Data: Qualitative variables like gender, blood type, etc

Numerical Data

Definition of Numerical Data:

- Numerical data involves measurable quantities and has a sense of measurement attached to it.
- Examples of numerical data:
  - Person's age, height, weight, blood pressure, heart rate, temperature, number of teeth, number of bones, number of family members.
- Also referred to as quantitative data in statistics.
- Types of Numerical Data:
  - Numerical data can be broadly classified into two types: Discrete Data and Continuous Data.
    - Discrete Data:
      - Data that is countable and whose values can be listed out.
      - Values are finite and distinct.
      - Examples:
        - Number of heads in 200 coin flips (values from 0 to 200).
        - Number of students in a class.
      - Discrete Variable: A variable that can only take distinct, countable values.
        - Example: The Rank of a student in a classroom (1, 2, 3, etc.).
        - Example: The Country a person belongs to (Nepal, India, Norway, Japan).

Numerical Data

- Continuous Data:
  - Data that can take an infinite number of values within a specific range.
  - The values are not countable and can be measured with higher precision.
  - Examples:
    - Temperature in a city (can have infinite decimal values within a specific range).
    - Weight (can have infinite possible values like 50.1 kg, 50.12 kg, etc.).
  - Continuous Variable: A variable that can take any numerical value within a range.
  - Example: The Temperature of a city today, which could have infinitely many possible values.

Categorical Data

- Categorical data represents the characteristics of an object or phenomenon.
- It involves non-numeric information and is often referred to as qualitative data in statistics.
- Examples of categorical data:
  - Gender: Male, Female, Other, Unknown
  - Marital Status: Annulled, Divorced, Interlocutory, Legally Separated, Married, Polygamous, Never Married, Domestic Partner, Unmarried, Widowed, Unknown
  - Movie Genres: Action, Adventure, Comedy, Crime, Drama, Fantasy, Historical, Horror, Mystery, Philosophical, Political, Romance, Saga, Satire, Science Fiction, Social, Thriller, Urban, Western
  - Blood Type: A, B, AB, O
  - Types of Drugs: Stimulants, Depressants, Hallucinogens, Dissociatives, Opioids, Inhalants, Cannabis
- Categorical Variable:
  - A variable that describes categorical data is called a categorical variable.
  - Categorical variables can take one of a limited number of values or categories.
- Types of Categorical Variables:
  - Binary Categorical Variable:
    - A variable that can take exactly two values.
    - Also referred to as a dichotomous variable.
    - Example: Experiment results (Success, Failure).

Categorical Data

- Polytomous Variable:
  - A variable that can take more than two values.
  - Example: Marital Status (Annulled, Divorced, Married, Widowed, etc.).
- Measurement Scales for Categorical Data:
  - Most categorical datasets follow either Nominal or Ordinal measurement scales.
- Understanding Nominal and Ordinal Scales:
  - Nominal Scale: Categorical variables with no inherent order (e.g., Gender, Movie Genres).
  - Ordinal Scale: Categorical variables with a defined order but no precise numeric difference between categories (e.g., Marital Status categories).

Measurement Scales

Introduction to Measurement Scales:

- There are four types of measurement scales in statistics: Nominal, Ordinal, Interval, and Ratio.
- These scales are essential in academic research and data analysis, as they help determine the nature of data and the appropriate statistical methods.

Nominal Scale:

- Used for labeling variables without any quantitative value.
- Key Characteristics:
  - Categories are mutually exclusive.
  - No numerical importance or order.
- Examples:
  - Gender: Male, Female, Third gender/Non-binary, Prefer not to answer
  - Languages spoken in a country
  - Biological species
  - Parts of speech in grammar (noun, verb, adjective)
  - Taxonomic ranks in biology (Archea, Bacteria, Eukarya)
- Data Type: Qualitative.

Important Note: Numbers used as labels in nominal measurement have no concrete numerical value or meaning. Arithmetic calculations cannot be performed.

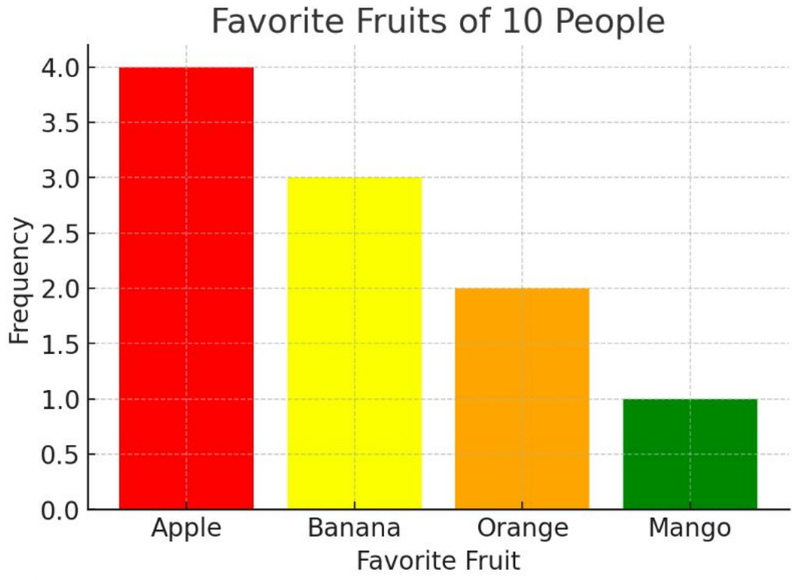
Measurement Scales

- Why Understanding Nominal Data is Important:
  - Frequency: Rate at which a label occurs in a dataset.
  - Proportion: Calculated by dividing the frequency by the total number of occurrences.
  - Visualization: Nominal data can be visualized using pie charts or bar charts.
- Why This Matters:
  - Understanding the type of data helps in:
    - Deciding which type of computation to perform.
    - Selecting the appropriate model to fit on the data.
    - Determining the visualization method.
- Example: For nominal data, pie charts or bar charts are ideal visualizations.

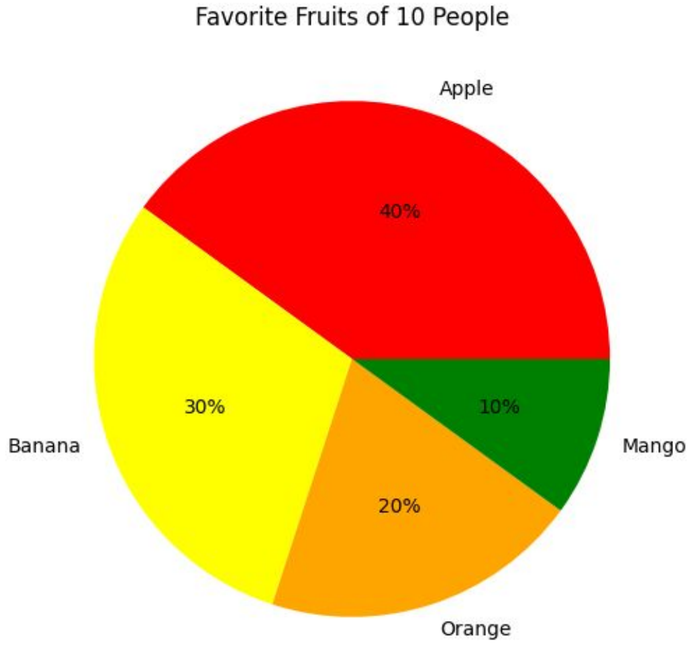
Nominal Scale - Example

Person	Fav Fruit
1	Apple
2	Banana
3	Apple
4	Orange
5	Banana
6	Mango
7	Apple
8	Orange
9	Banana
10	Apple

Fav. Fruit	Frequency
Apple	4
Banana	3
Orange	2
Mango	1



Fav. Fruit	Proportion
Apple	4/10 = 0.4 (40%)
Banana	3/10 = 0.3 (30%)
Orange	2/10 = 0.2 (20%)
Mango	1/10 = 0.1 (10%)

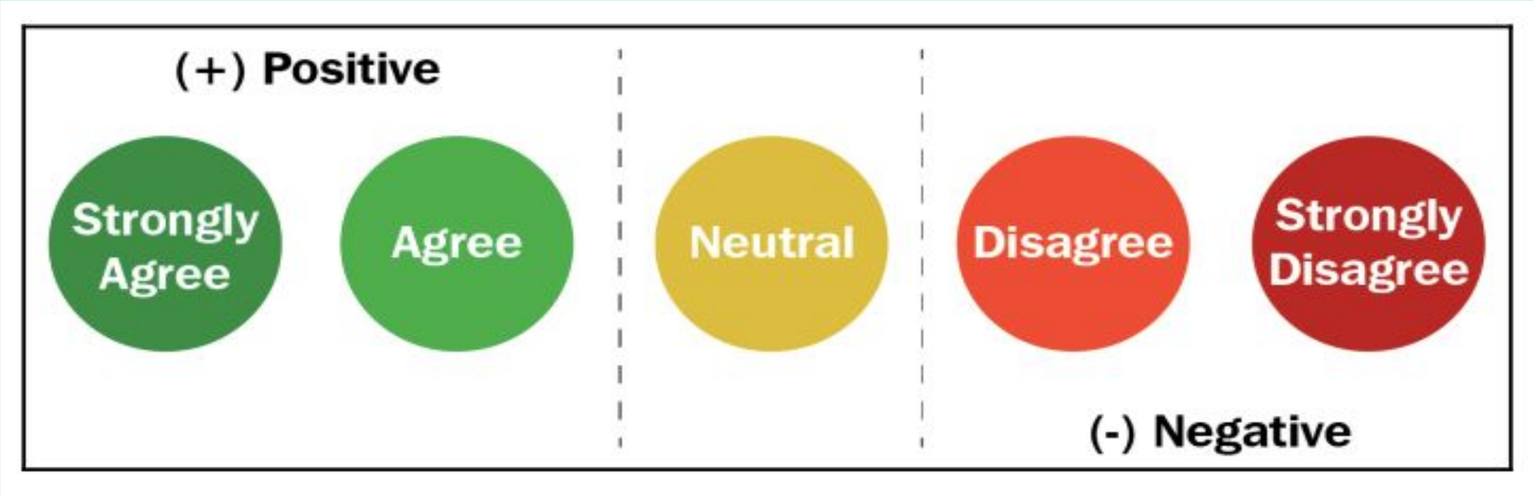


Ordinal Scale

The ordinal scale involves data that have a defined order, but the differences between the values are not meaningful.

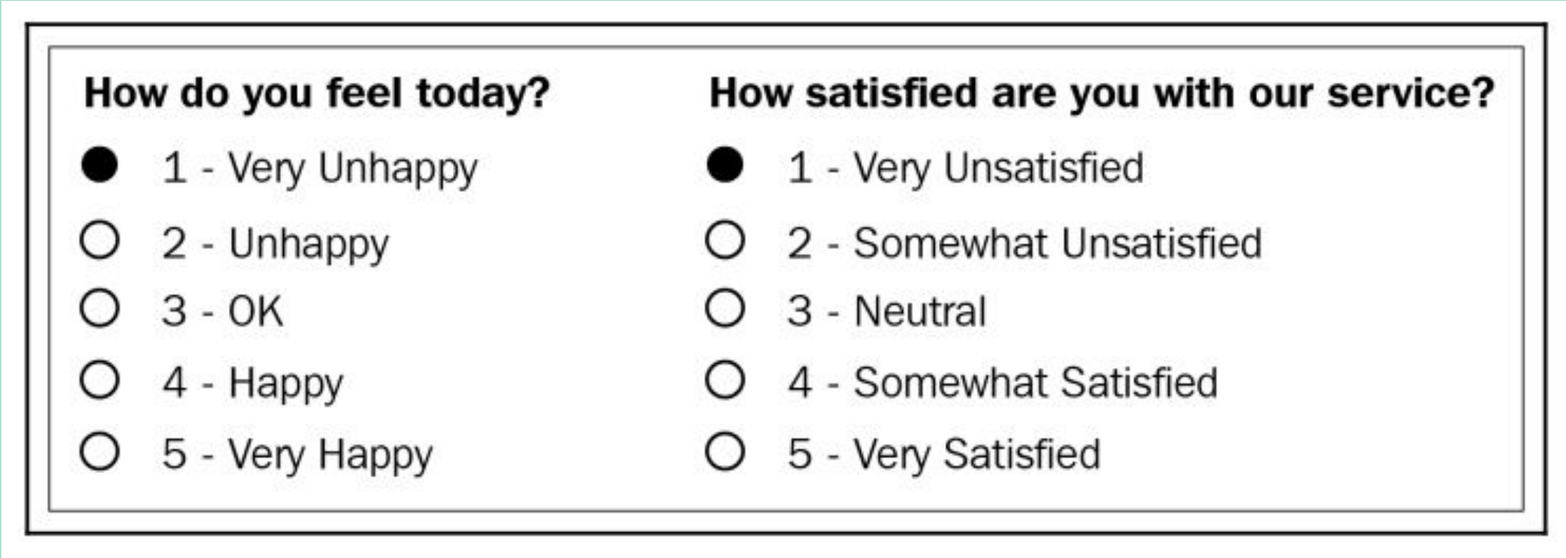
Key Characteristics:

- Order matters: The sequence of values represents a ranking or hierarchy.
- No meaningful difference between consecutive values (i.e., the difference between 1st and 2nd may not be the same as the difference between 3rd and 4th).
- Example: Likert Scale:
  - A common example of an ordinal scale is the Likert scale, used in surveys to measure people's opinion.



- These responses represent a ranked order, but the difference between, say, "Strongly Agree" and "Agree" isn't numerically defined.

Ordinal Scale



How do you feel today?

- ● 1 - Very Unhappy
- ○ 2 - Unhappy
- ○ 3 - OK
- ○ 4 - Happy
- ○ 5 - Very Happy

How satisfied are you with our service?

- ● 1 - Very Unsatisfied
- ○ 2 - Somewhat Unsatisfied
- ○ 3 - Neutral
- ○ 4 - Somewhat Satisfied
- ○ 5 - Very Satisfied

- Characteristics of Ordinal Data:
  - Order of Values: Values are ranked or ordered (1st, 2nd, 3rd, etc.).
  - Median: The median can be used to measure central tendency.
  - Mean: The average (mean) is not suitable, as the data is not uniformly spaced.
- Why Understanding Ordinal Data is Important:
  - Knowing that your data follows an ordinal scale helps you choose the right statistical analysis and visualization.
  - For ordinal data, you should avoid calculations like averages and focus on the median to understand central tendency.

Interval

- The interval scale has both order and exact differences between values, but it does not have an absolute zero.
- Key Characteristics:
  - Order: Values have a specific rank.
  - Exact differences: The difference between any two consecutive values is meaningful.
  - No Absolute Zero: The scale doesn't have a true zero point, meaning you cannot make ratios (e.g., you cannot say 20°C is twice as hot as 10°C).
- Examples:
  - Temperature (in Celsius or Fahrenheit): The difference between 10°C and 20°C is the same as between 20°C and 30°C, but 0°C does not represent the absence of temperature.
  - Calendar Dates: The difference between January 1st and January 2nd is meaningful, but the zero point (start of the calendar) is arbitrary.
- Statistical Measures:
  - You can compute mean, median, and mode for interval data.
  - Standard deviation can also be calculated.

Ratio

- Definition:
  - The ratio scale has order, exact differences, and also includes an absolute zero. This allows for the full range of mathematical operations.
- Key Characteristics:
  - Order and Exact Differences: As in the interval scale, the values have a defined order, and the differences between them are meaningful.
  - Absolute Zero: There is a true zero, meaning zero indicates the complete absence of the measured attribute. This allows for meaningful ratios.
- Examples:
  - Mass: Zero mass means no mass at all.
  - Length: Zero length means no length at all.
  - Time Duration: Zero time means no elapsed time.
  - Energy, Volume, Electrical Current: Zero indicates complete absence of the quantity.
- Statistical Measures:
  - All basic operations (addition, subtraction, multiplication, division) are valid for ratio scales.
  - Mean, median, mode, and standard deviation can all be calculated.
  - Coefficient of Variation can also be computed.

Ratio

Provides:	Nominal	Ordinal	Interval	Ratio
The "order"of values is known		✓	✓	✓
"Counts," aka "Frequency of Distribution"	✓	✓	✓	✓
Mode	✓	✓	✓	✓
Median		✓	✓	✓
Mean			✓	✓
Can quantify the difference between each value			✓	✓
Can add or subtract values			✓	✓
Can multiple and divide values				✓
Has "true zero"				✓

Comparing EDA, Classical, and Bayesian Data Analysis

Classical Data Analysis

- Relies heavily on pre-defined models.
- Assumes data follows a specific distribution or pattern.
- Focus is on hypothesis testing and statistical inference.

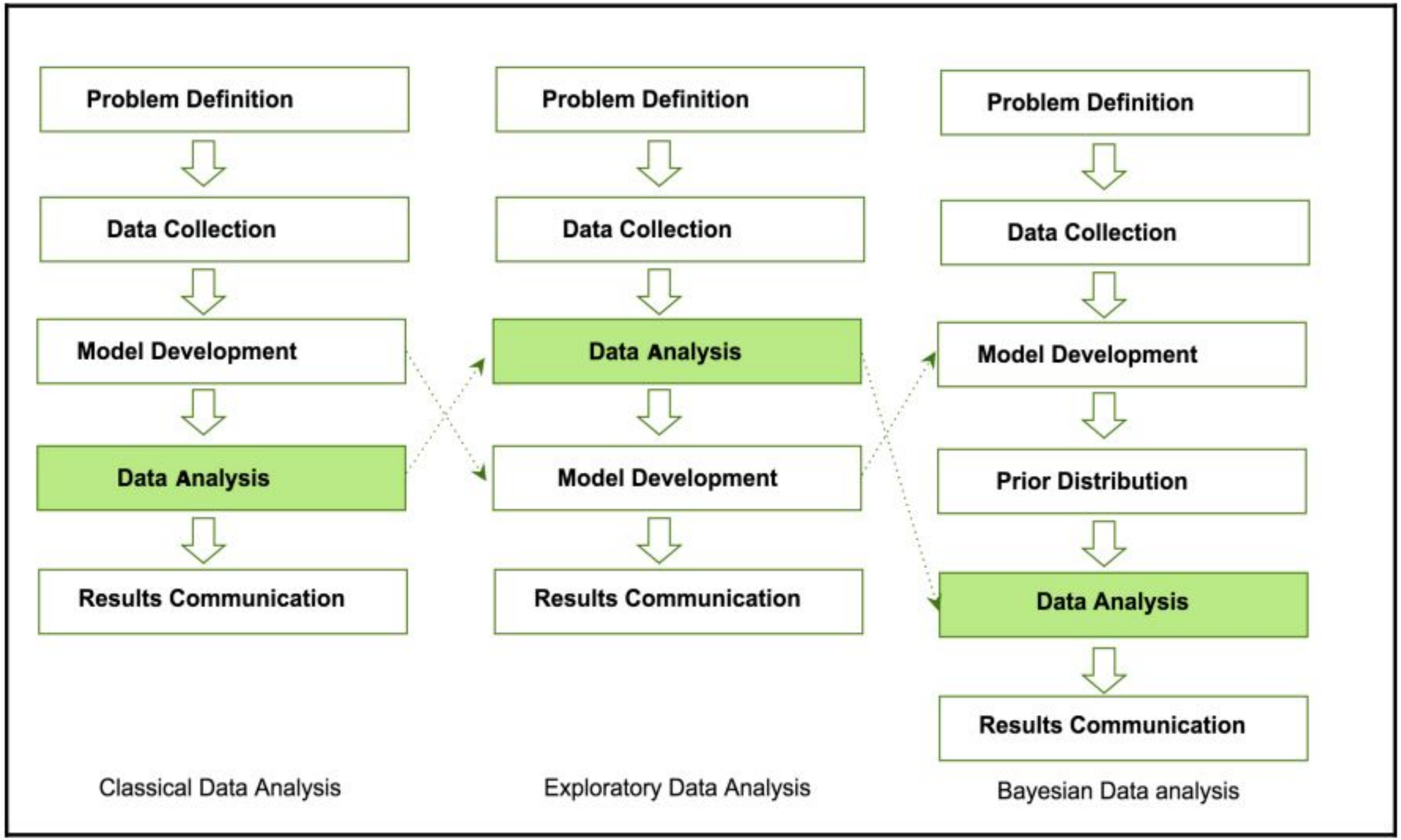
Exploratory Data Analysis (EDA)

- Focuses on understanding the data's structure, patterns, and anomalies.
- Uses visualizations (e.g., histograms, scatter plots) extensively.
- Avoids imposing deterministic or probabilistic models initially.
- Aims to explore data and generate hypotheses rather than confirm them.

Bayesian Data Analysis:

- Integrates prior beliefs (prior probability distributions) with the data.
- Focuses on updating these priors using evidence to compute the posterior probability distribution.
- Provides a probabilistic framework for decision-making.
- Useful in scenarios with limited data or when prior knowledge is crucial.

Comparing EDA with classical and Bayesian analysis



Getting started with EDA

Why Python is the main tool for data analysis?

- Popularity and Adoption
- Ease of Use
- Rich Ecosystem of Libraries
- Versatility
- Active Community Support
- Cross-Platform Compatibility

Essential Python Tools and Packages for Data Analysis

Python Programming Basics

- Core concepts: Variables, strings, and data types
- Logic flow: Conditionals and functions
- Data structures: Sequences, collections, and iterations
- File handling: Reading and writing files
- Object-Oriented Programming (OOP): Classes, objects, and methods

NumPy

- Array creation: Initialize and copy arrays, divide arrays
- Array operations: Perform mathematical and logical operations
- Advanced indexing: Array selection and manipulation
- Multi-dimensional arrays: Work with matrices and tensors
- Linear algebra: Use built-in linear algebraic functions
- NumPy functions: Leverage a wide range of pre-defined functions

pandas

- DataFrames: Create and manipulate DataFrame objects
- Data subsetting: Indexing and filtering data
- Arithmetic and mapping: Apply operations across datasets
- Index management: Optimize and control DataFrame indexing
- Visual analysis: Customize DataFrame styling for better insights

Essential Python Tools and Packages for Data Analysis

Matplotlib

- Plotting basics: Load and visualize linear datasets
- Customization: Adjust axes, grids, labels, titles, and legends
- Exporting plots: Save visualizations in various formats

SciPy

- Statistical analysis: Use SciPy's statistical packages
- Descriptive statistics: Summarize data distributions
- Inference and analysis: Perform hypothesis testing and data modeling